The Privacy-Preserving Big-Data Newsvendor: Efficiency of Noise Injection and Convolution Smoothing

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Privacy Concerns in Data-Driven Inventory Control

- Advances in machine learning have empowered firms to use data for better inventory decisions: Amazon (Spiegel et al. 2013); JD.com (Qi et al. 2022)
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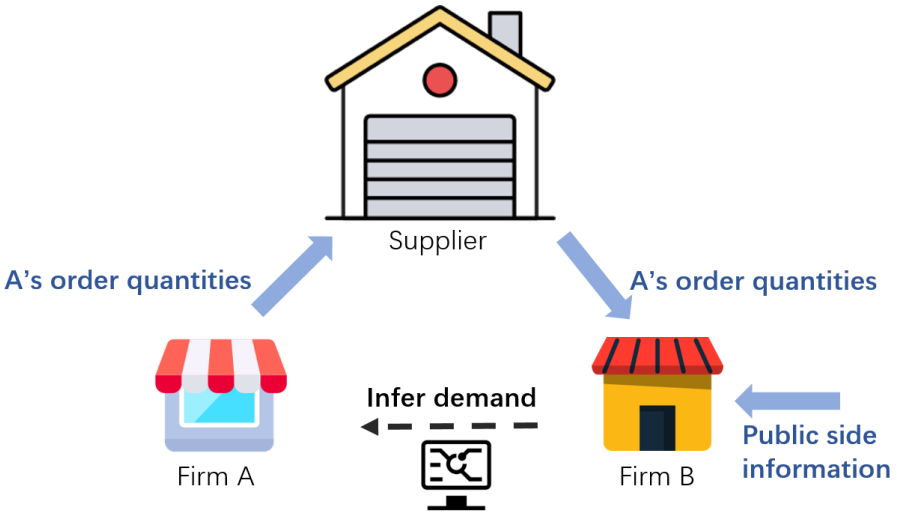
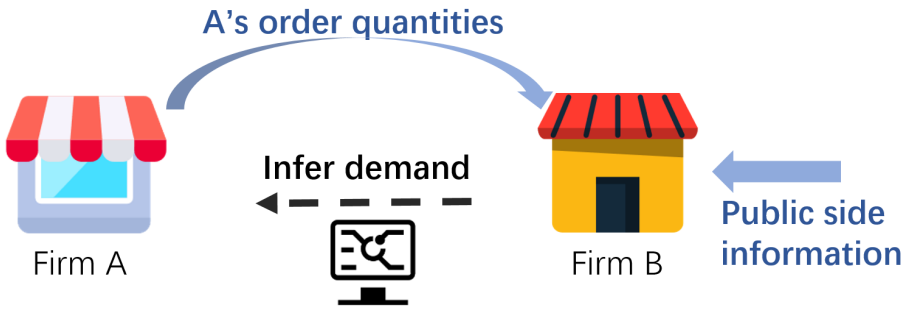
- A real case from Anand and Goyal (2009): Firm B bought A's order quantity info from the supplier for its strategical inventory planning and promotions, finally ruling out A from the market.



Motivating Example: the Data-Driven Newsvendor Problem



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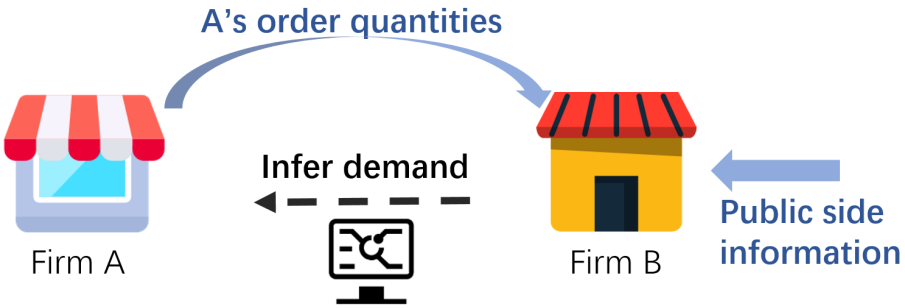

- Suppose firm A uses a single-item single-period data-driven newsvendor model with critical ratio $r = 0.5$.
- Binary demand $y \in \{0, 1\}$. (public side info)
- Suppose that firm A has run the business for five days and observed 5 historical demand:

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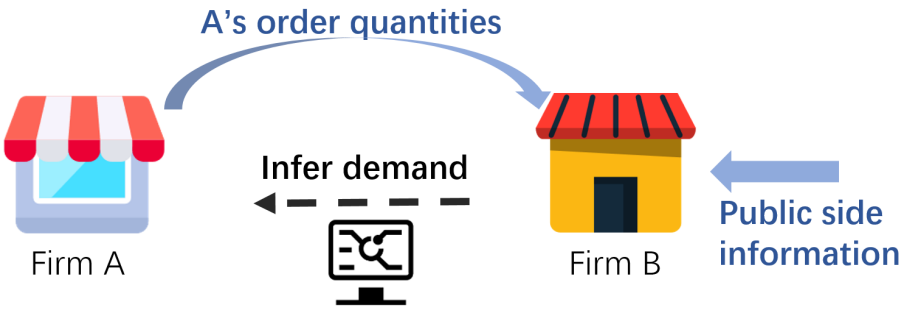

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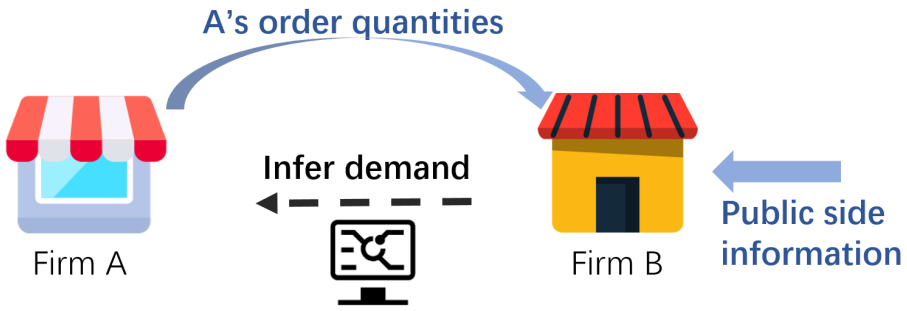

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$$\implies \text{optimal } \widehat{q}_6 = \widehat{F}^{-1}(r) = \inf_{y\in\mathbb{R}}\left\{y : \widehat{F}(y) \ge r\right\} = 1. \qquad (\widehat{q}_n \xrightarrow{P} q^*)$$

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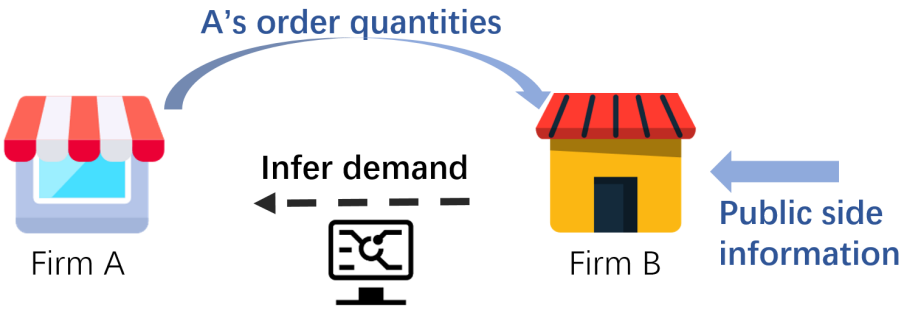

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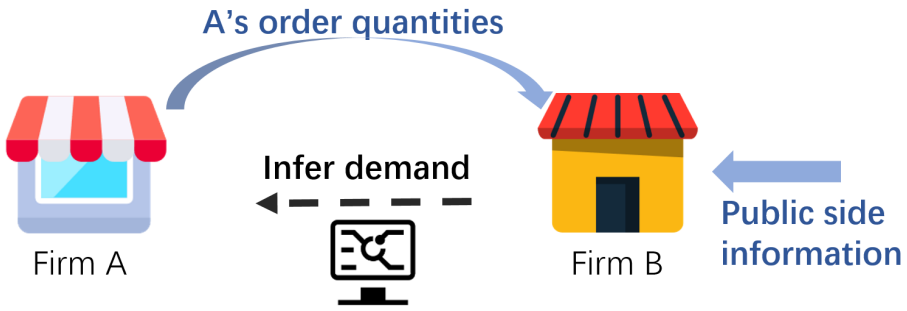

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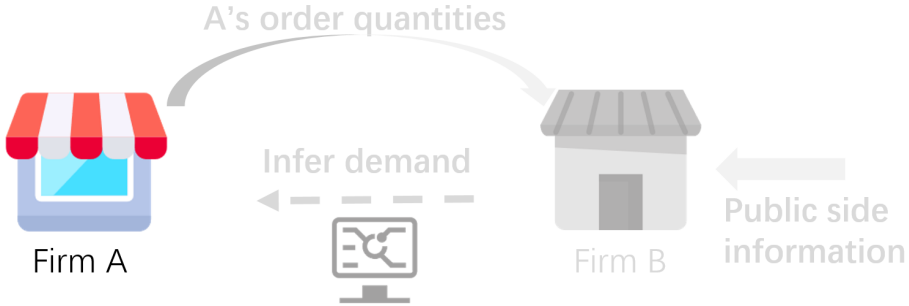
  - If $y_6 = 0 \implies$ optimal $\widehat{q}_7 = \widehat{F}^{-1}(r) = 0.$
- Firm B can accurately infer $y_6 = 0$ by comparing order quantities.

Research Questions



Research Questions:

- Can the newsvendor (firm A) **secure privacy** so that an adversary cannot infer any historical demand in the dataset by analyzing observed order quantities?
- What is the **trade-off** between privacy protection and decision optimality (in asymptotic regime)?

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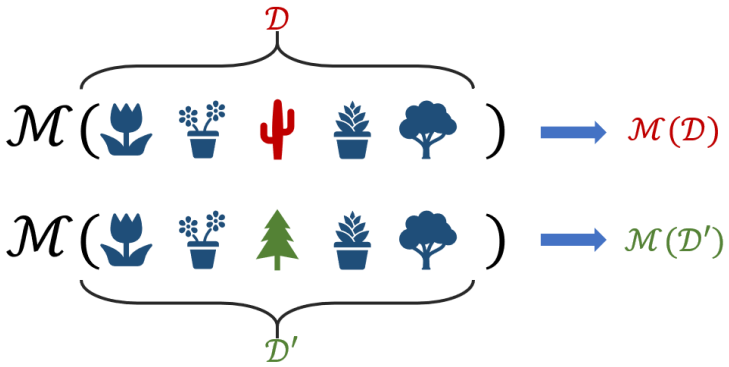
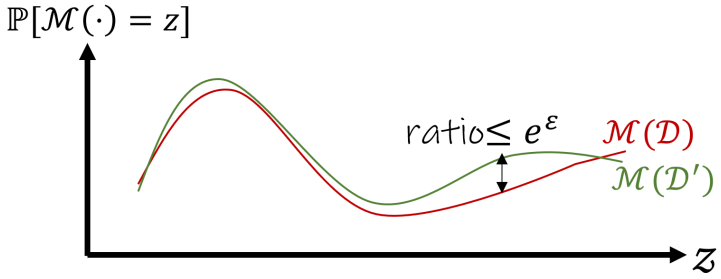
- A randomized algorithm $\mathcal{M} : \mathcal{X}^n \times \mathcal{Y}^n \to \mathcal{Z}$ is ε-differentially private (ε-DP) if for any neighboring datasets $\mathcal{D} \sim \mathcal{D}'$ that differ in only one record , $\frac{\mathbb{P}[\mathcal{M}(\mathcal{D})\in\mathcal{S}]}{\mathbb{P}[\mathcal{M}(\mathcal{D}')\in\mathcal{S}]} \leq e^{\varepsilon}, \forall \mathcal{S} \subseteq \mathcal{Z}$. (Smaller $\varepsilon > 0$, higher privacy guarantee, lower privacy loss)





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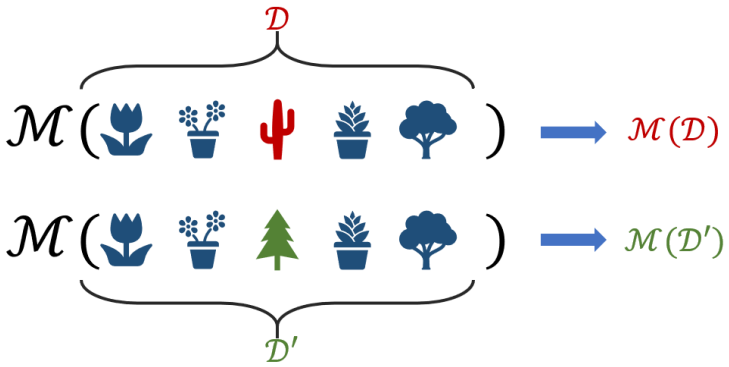

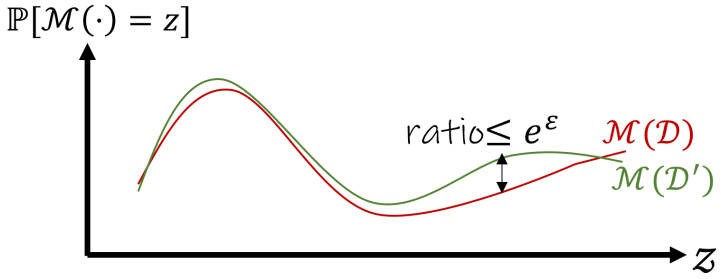

- We will adopt a **weaker** version: algorithm $\mathcal{M}$ is (ε, δ)-DP, if

$$\mathbb{P}\left[\mathcal{M}(\mathcal{D}) \in \mathcal{S}\right] \leq e^{\varepsilon} \cdot \mathbb{P}\left[\mathcal{M}(\mathcal{D}') \in \mathcal{S}\right] + \delta, \quad \forall \mathcal{D} \sim \mathcal{D}', \forall \mathcal{S} \subseteq \mathcal{Z},$$

where $\delta \in [0, 1]$ is a given parameter that controls the occurrence that the ratio$\geq e^{\varepsilon}$. (Smaller ε, δ, higher privacy guarantee, lower privacy loss)

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Our Work

We consider two relevant cases:

1. **Sampling-based (not covered in this talk)**: Demand $y \sim \mathbb{P}$ **unknown**, and the newsvendor has a dataset $\mathcal{D} = (y_1, \ldots, y_n) \in \mathcal{Y}^n$ of n *i.i.d* historical demand.

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2. **Feature-based (this talk)**: Demand $y = \langle \boldsymbol{\theta}^*, \boldsymbol{x} \rangle + \epsilon(\boldsymbol{x})$. The newsvendor has a dataset $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$ of n *i.i.d* data points from **unknown** $\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})$.

Related Literature

- Differential Privacy: Dwork et al. (2014), Vadhan (2017).
- Our work is technically related to **DP Stochastic Convex Optimization**:
  - Kifer et al. (2012), Bassily et al. (2014), Bassily et al. (2019), Feldman et al. (2020), Bassily et al. (2020), Asi et al. (2021), Bassily et al. (2021), etc.
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- Growing literature on DP Operations Management:
  - **Pricing**: Chen et al. (2022a), Chen et al. (2022b), Lei et al. (2020).
  - **Recommendation**: Fu et al. (2022), Küçükgül et al. (2022).
  - **Inventory Control**: Our work.

Differentially Private Newsvendor Problem (DP-NV)

- Newsvendor cost $c(u) = ru^+ + (1-r)(-u)^+, \forall u \in \mathbb{R}$.
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- The (ε, δ)-differentially private newsvendor problem (DP-NV) is modelled as

$$\inf_{\widehat{\boldsymbol{\theta}}^\pi \in \mathscr{F}_{\varepsilon,\delta}} \left\{ \sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\pi; \mathbb{P}) \right\},$$

  - $\mathcal{R}(\pi; \mathbb{P}) := \mathbb{E}_{\mathcal{D},\pi}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}^\pi; \mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*; \mathbb{P})$ is the excess generalization risk of $\widehat{\boldsymbol{\theta}}^\pi$;
  - $\mathscr{F}_{\varepsilon,\delta}$ is the family of all (ε, δ)-DP mappings from $\mathcal{X}^n \times \mathcal{Y}^n \to \mathbb{R}^d$;
  - $\mathcal{P}(\mathcal{X} \times \mathcal{Y})$ is the class of all probability measures over $\mathcal{X} \times \mathcal{Y}$.
- Propose some private algorithms instead of analytically solving it.

Noise Injection Ensures DP but Fails in Optimality

Let f be a function of interest, and $f(\mathcal{D})$ be the true data-driven decision.

- Laplace Mechanism ensures ε-DP: inject a Laplace noise and obtain a noisy decision:

$$\mathcal{M}(\mathcal{D}) := f(\mathcal{D}) + Lap(\mathrm{GS}_f/\varepsilon),$$

where $\mathrm{GS}_f := \sup_{\mathcal{D}\sim\mathcal{D}'} \|f(\mathcal{D}) - f(\mathcal{D}')\|_1$.

  - Lap(b) PDF: $p(x; b) = \frac{1}{2b} e^{-\frac{x}{b}}$.
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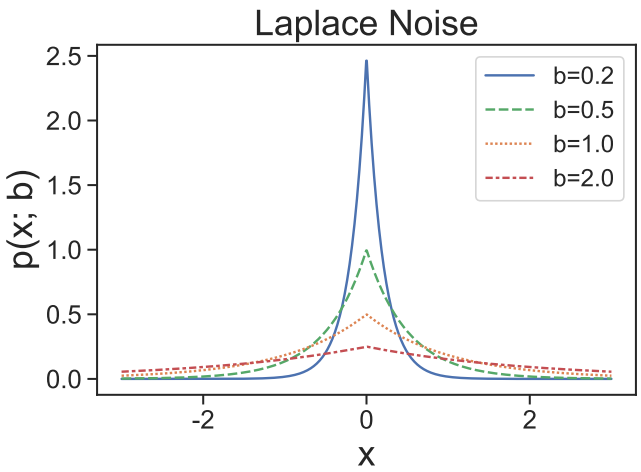

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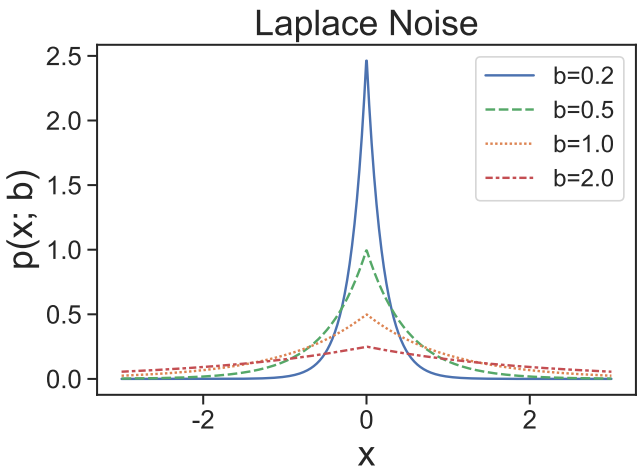


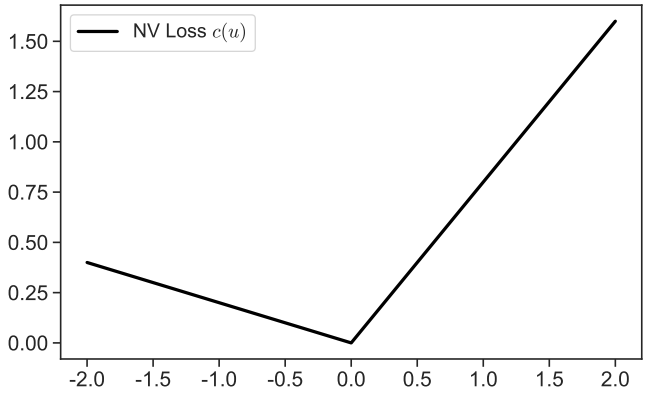
The Failure of Noise Injection in Decision Optimality

For any $\varepsilon > 0$, let $\widehat{q}^l = \widehat{q} + Lap(\mathrm{GS}_{\widehat{q}}/\varepsilon)$. Then, $\widehat{q}^l$ is ε-DP, but for any $\epsilon \in (0, \max\{q^*, \bar{y} - q^*\})$, there exists an optimality gap:

$$\Pr\left[\left|\widehat{q}^l - q^*\right| \geq \epsilon\right] \geq \frac{1}{2}\exp\left(-\frac{(q^* + \epsilon)\varepsilon}{\bar{y}}\right) + \frac{1}{2}\exp\left(-\frac{(\bar{y} + \epsilon - q^*)\varepsilon}{\bar{y}}\right).$$

Convolution Smoothing

- **Technical Challenge**: $c(\cdot)$ is nonsmooth
  $\Longrightarrow$ estimator $\widehat{\boldsymbol{\theta}}$ is unstable to input dataset
  $\Longrightarrow$ large injected noise
  $\Longrightarrow$ failure of noise injection



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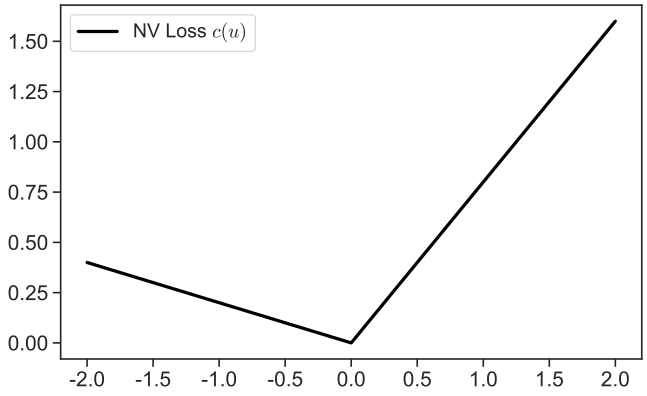
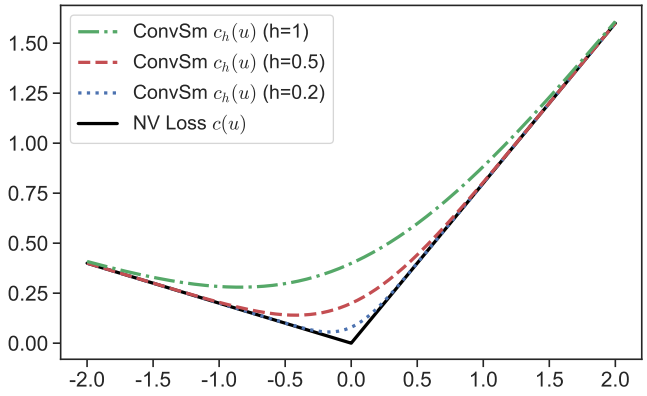


- **Convolution Smoothing** (weighted average over neighbors), bandwidth $h > 0$:

$$c_h(u) := \int_{-\infty}^{\infty} c(v) \cdot K((u-v)/h)/h \, dv$$

where $K : \mathbb{R} \to \mathbb{R}_+$ is a nonnegative **Kernel function** (or weight function) that satisfies some regularity conditions.



Private Algorithm 1: Gradient Perturbation (GP)

- Denote $\ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) := c_h(y - \boldsymbol{\theta}^\top \boldsymbol{x})$
- **Main idea**: Solve an empirical risk minimization problem

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \widehat{\mathcal{L}}_h(\boldsymbol{\theta}; \mathcal{D}) := \frac{1}{n} \sum_{i=1}^{n} \ell_h(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i),$$

with fixed-step Gradient Descent, but perturb gradients in each iteration.

Algorithm 3 Gradient Perturbation Policy (GP)

Parameters: privacy parameters $\varepsilon > 0, \delta > 0$, kernel function K with bandwidth $h > 0$, # itertaions $T = n^2$, noise variance $\sigma^2 = \frac{64TL^2 \ln(2.5T/\delta) \ln(2/\delta)}{n^2 \varepsilon^2}$, step size η.

Input: dataset $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n \in \mathcal{X}^n \times \mathcal{Y}^n$; an upcoming feature vector $\boldsymbol{x}_{n+1}$

Output: order quantity

1: Set initial point $\widehat{\boldsymbol{\theta}}_{h,1}^{\mathsf{GP}} = \mathbf{0}$
2: **for** $t = 1$ to $T - 1$ **do**
3: Sample a noise vector $\boldsymbol{w}_t \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{d \times d})$
4: Update $\widehat{\boldsymbol{\theta}}_{h,t+1}^{\mathsf{GP}} \leftarrow \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}} - \eta \left(\boxed{\nabla \widehat{\mathcal{L}}_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}; \mathcal{D}) + \boldsymbol{w}_t} \right)$
5: $\widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}} \leftarrow \frac{1}{T} \sum_{t=1}^{T} \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{GP}}$
6: Output order quantity $\langle \widehat{\boldsymbol{\theta}}_h^{\mathsf{GP}}, \boldsymbol{x}_{n+1} \rangle$

- Algorithm GP is (ε, δ)-DP. Proof by DP Composition Theory.

Private Algorithm 1: Gradient Perturbation (GP)

Theorem 1 (Performance Guarantee of Algorithm GP)

With appropriate step size η and bandwidth h, we have

$$\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{GP};\mathbb{P}) \leq \tilde{O}\left(\max\left\{\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n}}\right\}\right),$$

where $\mathcal{R}(\mathrm{GP};\mathbb{P}) := \mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})$ *is the excess generalization risk of Algorithm* GP *under distribution* $\mathbb{P}$*;*

- **Proof sketch**: uniform stability + shrinking empirical risk minimization
- **Insight**: $\mathcal{R}(\mathrm{GP};\mathbb{P}) = \mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right] + \mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})$

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$$\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{GP};\mathbb{P}) \leq \tilde{O}\left(\max\left\{\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n}}\right\}\right),$$

where $\mathcal{R}(\mathrm{GP};\mathbb{P}) := \mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})$ *is the excess generalization risk of Algorithm* GP *under distribution* $\mathbb{P}$*;*

- **Proof sketch**: uniform stability + shrinking empirical risk minimization
- **Insight:** $\mathcal{R}(\mathrm{GP};\mathbb{P}) = \underbrace{\mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right]}_{\text{privacy-induced error}} + \underbrace{\mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})}_{\text{statistical estimation error}}$

Private Algorithm 1: Gradient Perturbation (GP)

Theorem 1 (Performance Guarantee of Algorithm GP)

With appropriate step size η and bandwidth h, we have

$$\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{GP};\mathbb{P}) \leq \tilde{O}\left(\max\left\{\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n}}\right\}\right),$$

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- **Proof sketch**: uniform stability + shrinking empirical risk minimization
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Private Algorithm 1: Gradient Perturbation (GP)

Theorem 1 (Performance Guarantee of Algorithm GP)

With appropriate step size η and bandwidth h, we have

$$\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{GP};\mathbb{P}) \leq \tilde{O}\left(\max\left\{\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n}}\right\}\right),$$

where $\mathcal{R}(\mathrm{GP};\mathbb{P}) := \mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})$ *is the excess generalization risk of Algorithm* GP *under distribution* $\mathbb{P}$*;*

- **Proof sketch**: uniform stability + shrinking empirical risk minimization
- **Insight:** $\mathcal{R}(\mathrm{GP};\mathbb{P}) = \underbrace{\mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right]}_{\text{privacy-induced error} \leq \tilde{O}(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon})} + \underbrace{\mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})}_{\text{statistical estimation error} \leq O(\frac{1}{\sqrt{n}})}$
  - "privacy for free": when $n \gtrsim \frac{d\ln(1/\delta)}{\varepsilon^2}$, stat error dominates privacy error.

Private Algorithm 1: Gradient Perturbation (GP)

Theorem 1 (Performance Guarantee of Algorithm GP)

With appropriate step size η and bandwidth h, we have

$$\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{GP};\mathbb{P}) \leq \tilde{O}\left(\max\left\{\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n}}\right\}\right),$$

where $\mathcal{R}(\mathrm{GP};\mathbb{P}) := \mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})$ *is the excess generalization risk of Algorithm* GP *under distribution* $\mathbb{P}$*;*

- **Proof sketch**: uniform stability + shrinking empirical risk minimization
- **Insight:** $\mathcal{R}(\mathrm{GP};\mathbb{P}) = \underbrace{\mathbb{E}_{\mathcal{D},\mathrm{GP}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}}_h^{\mathrm{GP}};\mathbb{P})\right] - \mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right]}_{\text{privacy-induced error} \leq \tilde{O}(\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon})} + \underbrace{\mathbb{E}_{\mathcal{D}}\left[\mathcal{L}(\widehat{\boldsymbol{\theta}};\mathbb{P})\right] - \mathcal{L}(\boldsymbol{\theta}^*;\mathbb{P})}_{\text{statistical estimation error} \leq O(\frac{1}{\sqrt{n}})}$
  - "privacy for free": when $n \gtrsim \frac{d\ln(1/\delta)}{\varepsilon^2}$, stat error dominates privacy error.
- **Minimax risk**:

$$\Omega\left(\max\left\{\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n}}\right\}\right) \leq \inf_{\widehat{\boldsymbol{\theta}}^{\pi}\in\mathscr{F}_{\varepsilon,\delta}}\left\{\sup_{\mathbb{P}\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\pi;\mathbb{P})\right\}$$

$\Longrightarrow$ our algorithm GP is **statistically efficient**.

Private Algorithm 2: Stochastic Gradient Perturbation (SGP)

Algorithm 4 Stochastic Gradient Perturbation Policy (SGP)

Parameters: privacy parameters $\varepsilon > 0, \delta > 0$, kernel function K with bandwidth $h > 0$, # itertaions $T = n^2$, noise variance $\sigma^2 = \frac{32TL^2 \ln(n/\delta) \ln(1/\delta)}{n^2 \varepsilon^2}$, step size η.

Input: dataset $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n \in \mathcal{X}^n \times \mathcal{Y}^n$; an upcoming feature vector $\boldsymbol{x}_{n+1}$

Output: order quantity

Set initial point $\widehat{\boldsymbol{\theta}}_{h,1}^{\mathsf{SGP}} = \mathbf{0}$
for $t = 1$ to $T - 1$ **do**
  Uniformly sample a record $(\boldsymbol{x}_{(t)}, y_{(t)})$ from $\mathcal{D}$ with replacement
  Sample a noise vector $\boldsymbol{w}_t \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{d \times d})$
  Update $\widehat{\boldsymbol{\theta}}_{h,t+1}^{\mathsf{SGP}} \leftarrow \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}} - \eta \left(\boxed{\nabla \ell_h(\widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}; \boldsymbol{x}_{(t)}, y_{(t)}) + \boldsymbol{w}_t} \right)$
$\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}} \leftarrow \frac{1}{T} \sum_{t=1}^T \widehat{\boldsymbol{\theta}}_{h,t}^{\mathsf{SGP}}$
Output order quantity $\widehat{\boldsymbol{\theta}}_h^{\mathsf{SGP}\top} \boldsymbol{x}_{n+1}$

Theorem 2 (Privacy & Performance Guarantee of Algorithm SGP)

Algorithm SGP *is* (ε, δ)*-DP. Moreover, with appropriate step size* η *and bandwidth* h*, we have*

$$\sup_{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Y})} \mathcal{R}(\mathsf{SGP}; \mathbb{P}) \leq \tilde{O}\left(\max\left\{ \frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n}} \right\} \right).$$

Private Algorithm 3: Objective Perturbation (OP)

- **Main idea**: Solve a regularized empirical risk minimization model

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \widehat{\mathcal{L}}_h^{\mathrm{OP}}(\boldsymbol{\theta};\mathcal{D}) := \frac{1}{n}\sum_{i=1}^{n} \ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i) + \lambda\,\|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top\boldsymbol{\theta}}{n},$$

but perturb the objective function with Gaussian noise $\boldsymbol{b}\sim\mathcal{N}(0,\sigma^2\mathbf{I}_{d\times d})$, where $\sigma^2 = 8\ln(1/\delta)/\varepsilon^2$

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Theorem 3 (Privacy & Performance Guarantee of Algorithm OP)

Algorithm OP *is* (ε,δ)*-DP; Moreover, with appropriate bandwidth* h*, we have*

$$\sup_{\mathbb{P}\in(\mathcal{X}\times\mathcal{Y})} \mathcal{R}(\mathrm{OP};\mathbb{P}) \le \tilde{O}\left(\max\left\{\frac{\sqrt{d\ln(1/\delta)}}{n\varepsilon}, \frac{1}{\sqrt{n\varepsilon}}\right\}\right)$$

- **Proof sketch**: uniform stability + shrinking empirical risk minimization

Simulations with Exogenous Error Term ϵ

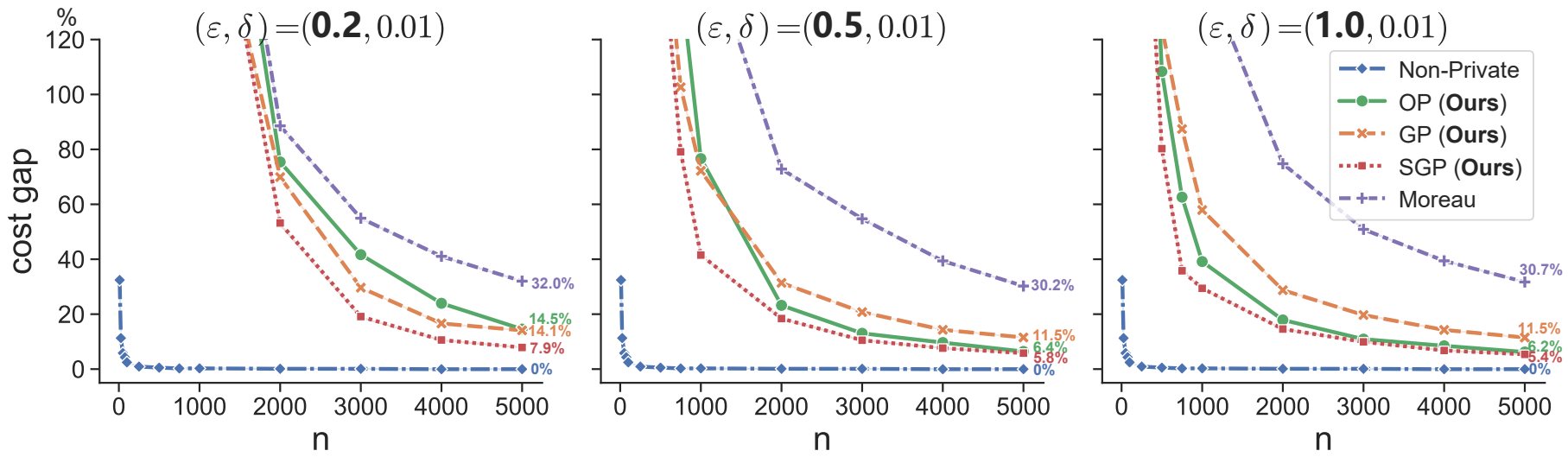


Figure: Cost gaps of corresponding policies against true optimal cost in percentage. Demand follows $y = 10 + 5x_1 - 2x_2 + \epsilon$, where $x_1 \sim \mathcal{N}(0, 2^2), x_2 \sim \mathcal{N}(0, 3^2)$, and $\epsilon \sim \mathcal{N}(0, 3^2)$

- Benchmarks:
  - "Non-Private": $\|\cdot\|_2^2$-regularized model, NV-ERM2 in Ban and Rudin (2019)
  - "Moreau": a private method by Bassily et al. (2019)

Simulations with Endogenous Error Term $\epsilon(\boldsymbol{x})$

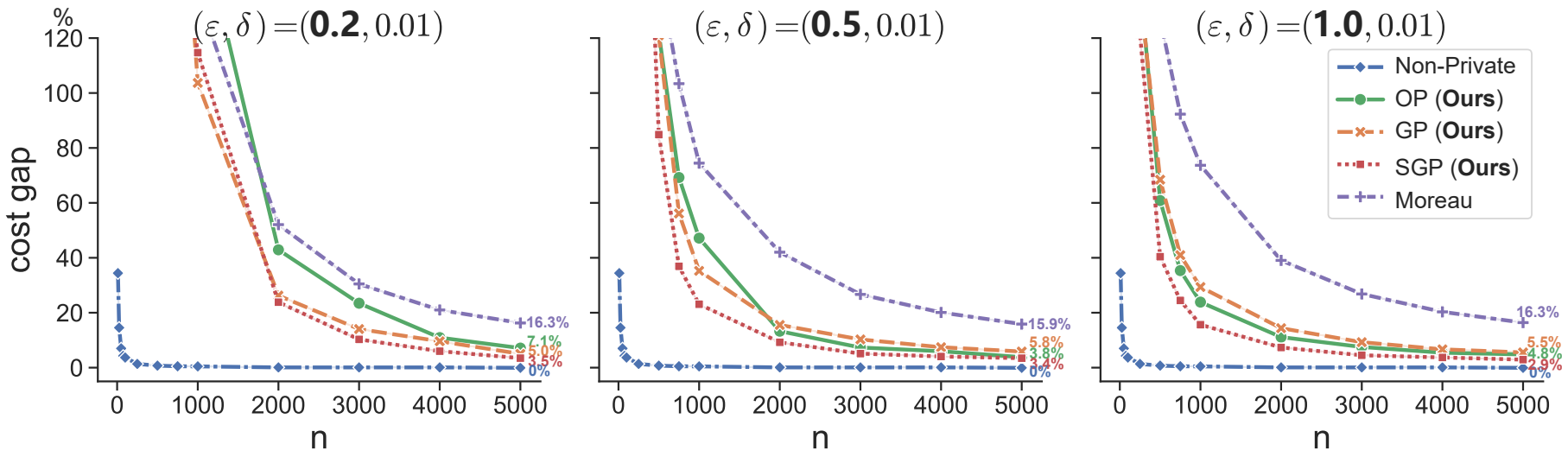
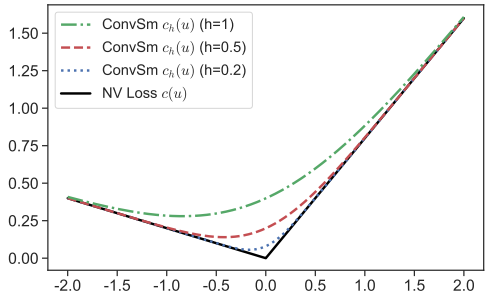


Figure: Same settings except that demand $y = 10 + 5x_1 - 2x_2 + \epsilon(\boldsymbol{x})$ with $\epsilon(\boldsymbol{x}) = \mathcal{N}(0, 9) + x_1 - x_2 + |x_1 x_2|$.

Conclusions and Takeaways

1. We considered a **privacy-preserving** data-driven newsvendor problem.
2. We employed **convolution smoothing** to smooth out the newsvendor function.



3. We proposed some private algorithms (**gradient perturbation; objective perturbation**), and characterized the **privacy-induced error** $\frac{\sqrt{d \ln(1/\delta)}}{n\varepsilon}$, which is rate optimal.
4. Our algorithms outperformed existing methods in simulations.

Thanks!

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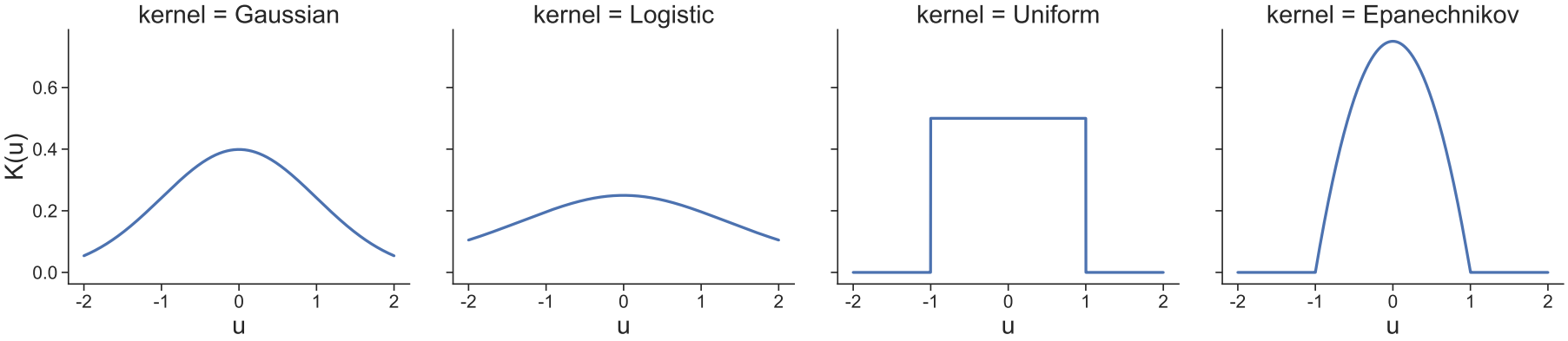
Kernel Functions

Conditions on Kernel Functions: Let $K : \mathbb{R} \to \mathbb{R}_+$ be a nonnegative function having following properties,

- **Integrate to 1**: $\int K = 1$;
- **Symmetry**: $K(u) = K(-u)$ for all $u \in \mathbb{R}$;
- **Monotonicity**: $K(u) \leq K(v)$, $\forall |u| \geq |v| \in \mathbb{R}$.
- **Finite absolute moments**: $\kappa_1 := \int_{-\infty}^{\infty} |u| \, K(u) \, du < \infty$, $\kappa_2 := \int_{-\infty}^{\infty} u^2 K(u) \, du < \infty$, and $\overline{K} := \sup_u K(u) < \infty$.



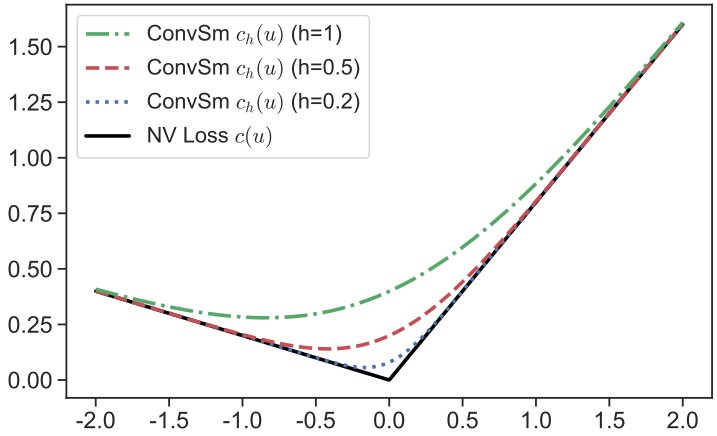
Properties of smoothed loss function

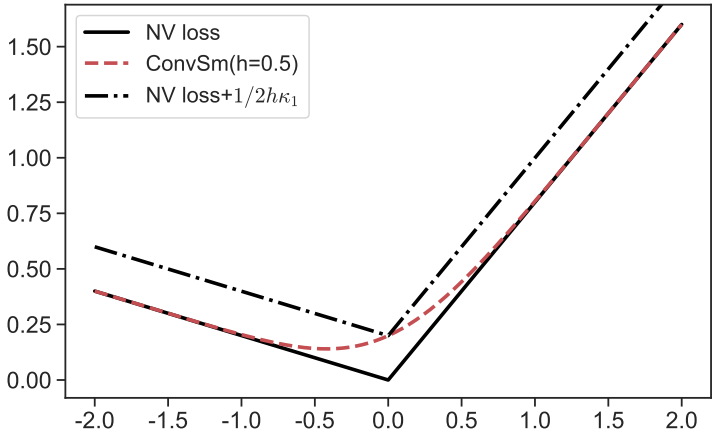
- Properties of $\ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) := c_h(y - \boldsymbol{\theta}^\top \boldsymbol{x})$:
  - **convex, twice-differentiable in $\boldsymbol{\theta}$**:

$$\nabla \ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) = \left[\mathcal{K}_h\left(\boldsymbol{\theta}^\top \boldsymbol{x} - y\right) - r\right] \cdot \boldsymbol{x}; \quad \nabla^2 \ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) = K_h\left(y - \boldsymbol{\theta}^\top \boldsymbol{x}\right) \cdot \boldsymbol{x}\boldsymbol{x}^\top \succcurlyeq \mathbf{0};$$

  - $\bar{r}B_x =: L$-**Lipschitz continuous** and $\overline{K}B_x^2/h =: \beta$-**smooth** in $\boldsymbol{\theta}$ w.r.t. $\|\cdot\|_2$;
  - **upper and lower bounded** by affine functions of ℓ.

$$\ell(\boldsymbol{\theta}; \boldsymbol{x}, y) \leq \ell_h(\boldsymbol{\theta}; \boldsymbol{x}, y) \leq \ell(\boldsymbol{\theta}; \boldsymbol{x}, y) + \frac{1}{2} h \kappa_1.$$





Potential Tractability Issue in Algo OP

The convex problem in OP is $\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \frac{1}{n}\sum_{i=1}^n \ell_h(\boldsymbol{\theta};\boldsymbol{x}_i,y_i) + \lambda\|\boldsymbol{\theta}\|_2^2 + \frac{\boldsymbol{b}^\top\boldsymbol{\theta}}{n}$

- Recall that

$$\ell_h(\boldsymbol{\theta};\boldsymbol{x},y) := c(y-\boldsymbol{\theta}^\top\boldsymbol{x})$$

$$c_h(u) = \frac{h}{2}\int_{-\infty}^{\infty}\left|\frac{u}{h}+v\right| K(v)\,dv + \left(r-\frac{1}{2}\right)u,$$

the tractability depends on representing the integral term $\int_{-\infty}^{\infty}\left|\frac{u}{h}+v\right| K(v)\,dv$.

Kernels	$K(u)$	$\int_{-\infty}^{\infty}\lvert u/h+v\rvert K(v)\,dv$
Gaussian	$\frac{1}{\sqrt{2\pi}}e^{\frac{-u^2}{2}}$	$\sqrt{\frac{2}{\pi}}e^{-\frac{u^2}{2h^2}} + u/h(1-2\Phi(-u/h))$
Logistic	$\frac{e^{-u}}{(1+e^{-u})^2}$	$u/h + 2\ln\left(1+e^{-u/h}\right)$
Uniform	$\frac{\mathbb{1}\{\lvert u\rvert\le 1\}}{2}$	$(\frac{u^2}{2h^2}+\frac{1}{2})\mathbb{1}\{\lvert u\rvert\le h\} + \left\lvert\frac{u}{h}\right\rvert\mathbb{1}\{\lvert u\rvert > h\}$
Epanechnikov	$\frac{3}{4}(1-u^2)\mathbb{1}\{\lvert u\rvert\le 1\}$	$(-\frac{u^4}{8h^4}+\frac{3u^2}{4h^2}+\frac{3}{8})\mathbb{1}\{\lvert u\rvert\le h\} + \left\lvert\frac{u}{h}\right\rvert\mathbb{1}\{\lvert u\rvert > h\}$

$$\left\{(u,t)\in\mathbb{R}^2 \middle| u+2\ln\left(1+e^{-u}\right)\le t\right\} = \left\{(u,t)\in\mathbb{R}^2 \middle| \begin{array}{l}\exists(v,x,y)\in\mathbb{R}^3: \\ x+y\le 1 \\ (x,1,-u-v)\in K_{exp} \\ (y,1,-v)\in K_{exp} \\ 2v\le t-u\end{array}\right\}$$

Estimator $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}(\boldsymbol{b})$ given by OP converges to $\boldsymbol{\theta}^*$

- The solution $\boldsymbol{\theta}_h^{\mathrm{OP}}(\boldsymbol{b})$ by OP is asymptotically optimal to $\boldsymbol{\theta}^*$ under some mild assumptions on data generating process:
  - (**Normalized $\boldsymbol{x}$**): $x_1 \equiv 1$, and $\mathbb{E}[x_i] = 0, \forall i = 2, \ldots, d$.
  - (**Lipschitz pdf**): For any $\boldsymbol{x} \in \mathcal{X}$, the pdf $f_{\epsilon|\boldsymbol{x}}$ of $\epsilon(\boldsymbol{x})$ exists, and is l_1-Lipschitz continuous.
  - (**Locally Strictly Positive Slope**): There exists a constant $\underline{f} > 0$ such that $f_{\epsilon|\boldsymbol{x}}(0) \geq \underline{f}$, for all $\boldsymbol{x} \in \mathcal{X}$.
  - (**Widely Spread $\boldsymbol{x}$**): The matrix $\boldsymbol{\Sigma} := \mathbb{E}_{\boldsymbol{x}}\left[\boldsymbol{x}\boldsymbol{x}^\top\right] \succ \mathbf{0}$ is positive definite, and has a minimal eigenvalue $\rho_1 > 0$.

Proposition 6 (Asymptotic Optimality of $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}(\boldsymbol{b})$)

If we set $\lambda = \frac{1}{B_\theta} \cdot \Theta\left(\sqrt{\frac{1}{n\varepsilon} + \frac{d\sigma^2}{n^2}}\right)$ *and* $h = \frac{\overline{K} B_x^2}{\lambda n \varepsilon}$ *as the same values in Algo.*OP*, and set privacy parameter* $\varepsilon > 0$ *and* $\delta \asymp n^{-w}$ *for some* $w > 0$*, then for any distribution* $\mathbb{P}$ *that satisfies above assumptions, with probability at least* $1 - \beta, \forall \beta \in (0, 1)$ *over the random draw of dataset* $\mathcal{D}$*, the private estimator* $\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}(\boldsymbol{b})$ *obtained from Algorithm* OP *satisfies*

$$\mathbb{E}_{\boldsymbol{b}}\left[\left\|\widehat{\boldsymbol{\theta}}_h^{\mathrm{OP}}(\boldsymbol{b}) - \boldsymbol{\theta}^*\right\|_2\right] \leq O\left(\max\left\{\sqrt{\frac{d + \ln(1/\beta)}{n}}, \sqrt{\frac{d \ln(1/\delta)}{n\varepsilon}}\right\}\right).$$

- **Proof sketch**: overall error = privacy-induced error + estimation error + approximation error.

Solving Time

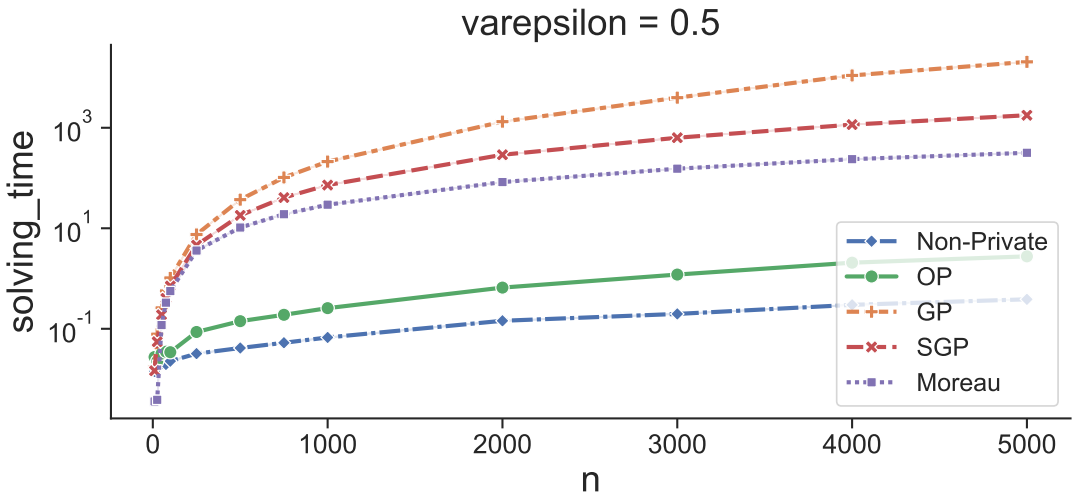


Figure: solving time in seconds

Table: Solving time (mean±std, in seconds)

Model	n=100	n=1000	n=5000
Non-Private	0.02±0.01	0.07±0.02	0.39±0.07
OP	0.03±0.01	0.26±0.02	2.76±0.23
GP	7.51±0.17	1323±10	20502±103
SGP	0.71±0.02	72.42±0.53	1775.11±7.74
Moreau	0.56±0.02	29.44±0.28	317.73±31.79